BRAC UNIVERSITY
Department of Computer Science and Engineering

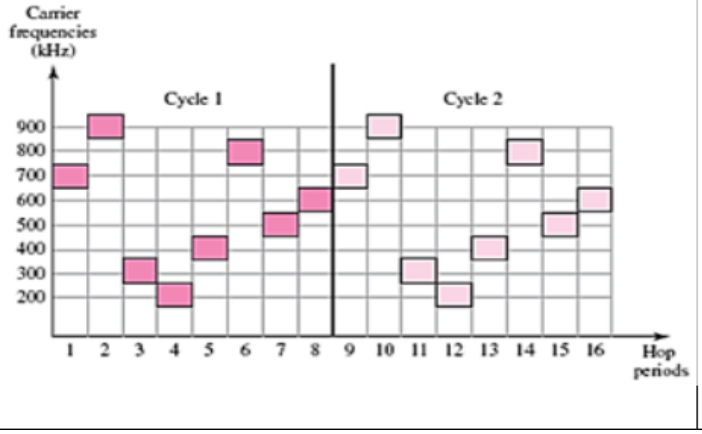
Examination: Semester Final | Semester: Spring 2024
Duration: 1 hr. 50 min | Full Marks: 50

CSE 320/EEE361/ECE361: Data Communications

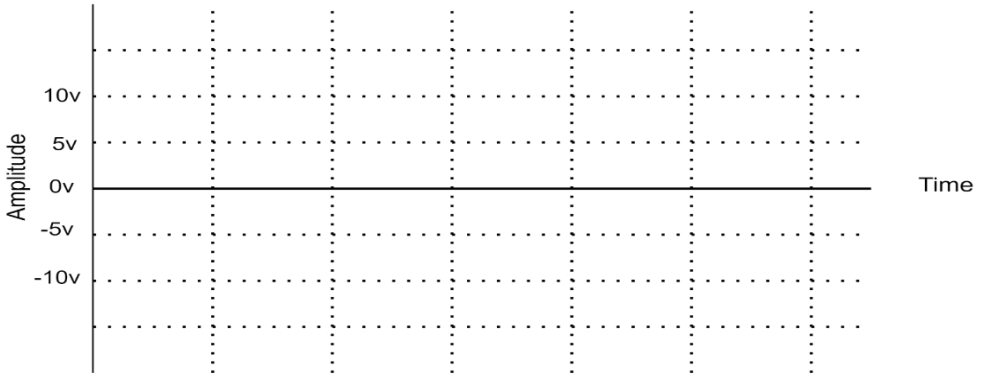
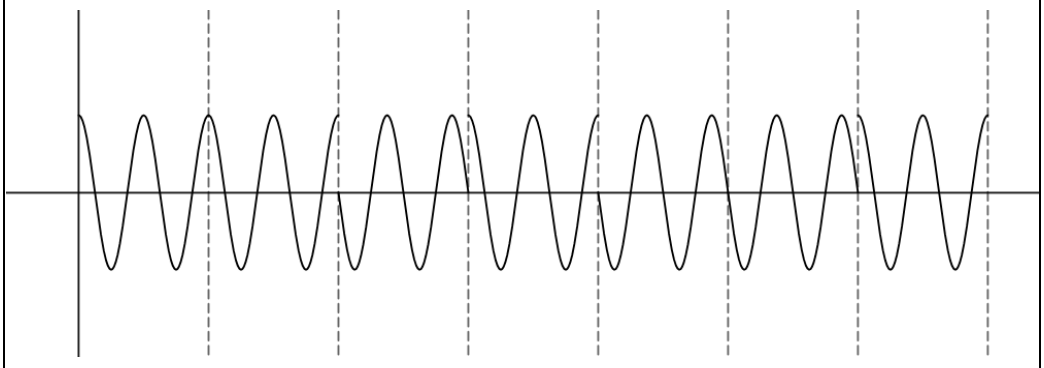
Answer the following questions.
Figures in the right margin indicate marks.

SET B

Name:	ID:	Section:

1. [CO3]	a)	Suppose 5 data sources with data rates of 72 Kbps, 144 Kbps, 144 Kbps, 72 Kbps and 288 Kbps are to be multiplexed with one synchronization bit. **Show** the visual representation of MUX with arbitrary input and output bit without adding extra pulses in the input channel.	[2]
[CO3]	b)	Suppose, each of the 5 channels send 6000 pages per minute where each page consists of 180 characters. If synchronous TDM is used to combine these sources using two-characters interleaving with one sync bit then answer the following questions: i. What is the input data rate (in bps) for each of the connections? ii. What is the input bit duration (in seconds)? iii. What is the frame rate (frames per second)? iv. What is the size of a frame in bits? v. What is the output data rate (in bps)? vi. What is the output bit duration (in seconds)?	[6]
[CO3]	c)	Suppose there are 5 hopping frequencies in FHSS. If one hopping frequency is used by station 1 currently, will any another station be able to use the same hopping frequency at some other period? Yes/no, explain briefly.  **Find** the pseudorandom generator's k-bit pattern output to match the given FHSS cycle and the Frequency hopping table.	[2+2]

k-bit	Frequency (kHz)
111	200
110	300
101	400
100	500
011	600
010	700
001	800
000	900

	d)	There is a digital medium with a data rate of 15 Mbps. How many 60-kbps voice channels can be carried by this medium if we use DSSS with the code sequence of '010110?	[2]
2. [CO2]	a)	**Differentiate** between coherent and non-coherent Frequency Shift Keying.	[2]
[CO2]	b)	In a Multi-level FSK, for each signal element, we want to send 2 bits at a time. We have used a carrier signal that has an amplitude of 10v and phase is 0 degree. If the frequency changes according to the following table, **draw** the modulated signal for the bit sequence 0110001101. ***[answer this question in the given graph below]*** Bit Pattern / # cycles of the signal element: 00 → 1; 01 → 2; 10 → 3; 11 → 4 	[5]
[CO2]	c)	**Determine** the digital bit stream from the analog signal below. The signal was modulated using Binary PSK where 0 means signal element with phase of π rad and 1 means signal element with phase of π/2 rad. 	[5]

3. **[CO4]**	a)	Suppose two channels are using the following linear block coding scheme for error detection and correction. In a linear block coding scheme, XORing two distinct valid codewords produces another distinct codeword. Dataword | Codeword 00 | 00000 01 | 01101 10 | ***?????*** 11 | 11011 i. From the table above, find the value of n and k for the notation C(n,k). ii. What would be the missing distinct codeword? iii. Can we successfully detect and correct up to 3 bits error using the scheme?	[5]
[CO4]	b)	Assume the following cases for a sender and a receiver device that use CRC for error control. Dataword: $x^6 + x^5 + x^3 + x^2$ Divisor: $x^4 + x^2 + 1$ **Show** the **sent codeword** in CRC **polynomial** format.	[5]
4. **[CO4]**	a)	A common multipoint link is being used by three stations. Station 1 is to the furthest left and Station 4 is to the furthest right, with the remaining stations in between. • Station 4 starts to send a frame to Station 1 at time t = 1.2 ms • The first bit of the frame reaches Stations 3, 2 and 1 at times t = 3.4 ms, t = 8.7 ms and 15.4 ms respectively • Station 4 is supposed to send the last bit of the frame at time t = 31.9 ms i. Sketch a space and time representation of the channel if it is using CSMA/CD. ii. What is the Transmission Time and Maximum Propagation Time from the given information? iii. What should be the Minimum Frame Size and does it meet the requirements given above if the bandwidth is 7Gbps?	[3+3+2]
[CO4]	b)	For CSMA/CA, why is the Time-out waiting period twice the propagation time?	[2]
[CO4]	c)	Is it possible to wait 12 slots of the contention window if the value of K is 2 in CSMA/CA? Explain your answer with a mathematical explanation. Mention the role of IFS in CSMA/CA.	[2+2]